

NM4QM 2026 · HANDS-ON SESSION · 45 MIN

Spin and anomalous Hall effects with WannierBerri and GPAW

Symmetry-adapted Wannier functions, spin–orbit coupling on demand, the spin Hall effect of graphene and the anomalous Hall effect of the altermagnet MnTe

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Two puzzles, two acts

0–8 min

Setup and ideas

8–19

Act 1: graphene

19–37

Act 2: MnTe

37–45

Your material,
questions

TWO PUZZLES

- Spin–orbit coupling turns graphene into a quantum spin Hall insulator with a gap of $42 \mu\text{eV}$
- MnTe has no net magnetisation, yet shows an anomalous Hall effect

WHAT YOU WILL DO

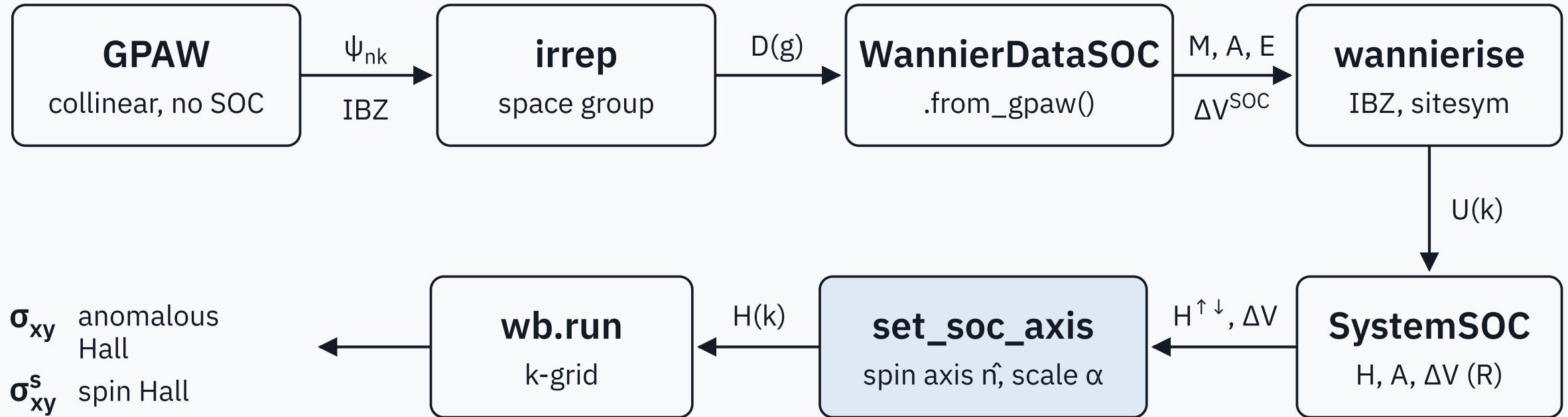
- Build symmetry-adapted Wannier functions straight from GPAW
- Switch spin–orbit coupling on, off and around, without new DFT
- Compute spin and anomalous Hall conductivities

Before we start: one check, 20 seconds

```
cd wannier-berri-demo && conda activate wbdemo
python setup/check_setup.py          # should end with "Ready."
cd 01_graphene
python 1_dft.py                      # 10 s
jupyter lab graphene.ipynb
```

- No environment yet: `bash setup/setup.sh` (15 min)
- DFT fails: `tar -xzf dft-files.tar.gz` in `wannier-berri-demo/`
- Fell behind: `bash run_all.sh` (6 min); finished notebooks in `reference/`
- Numbers on the slides are checkpoints: compare them with yours

One Python pipeline, GPAW to conductivities



Each box is one call in the notebooks. The shaded step needs no new DFT and no new Wannierisation.

ψ_{nk} : Bloch states · $D(g)$: symmetry matrices on them · M, A, E: overlaps, projections, energies · ΔV^{SOC} : SOC matrix elements from the PAW spheres · $U(k)$: Wannier gauge · $H, A, \Delta V(\mathbf{R})$: Hamiltonian, Berry connection and SOC in the Wannier basis · $\hat{n}, \alpha$: spin axis, SOC scale · σ_{xy} : anomalous Hall (charge current), σ_{xy}^s : spin Hall (spin current) conductivity

Berry curvature needs millions of k-points; DFT gives tens

Intrinsic anomalous Hall conductivity:

$$\sigma_{xy} = -\frac{e^2}{\hbar} \int_{\text{BZ}} \frac{d^3k}{(2\pi)^3} \sum_n f(\varepsilon_{n\mathbf{k}}) \Omega_n^z(\mathbf{k})$$

$$\Omega_n^z = -2 \text{Im} \sum_{m \neq n} \frac{v_{nm}^x v_{mn}^y}{(\varepsilon_n - \varepsilon_m)^2}$$

f : occupation, ε_n : band energy, $v_{nm}^a = \langle u_n | \partial_{k_a} H | u_m \rangle$. Ω_n^z is large and narrow wherever two bands come close.

Wannier functions give $H(\mathbf{k})$ at any $\mathbf{k}$:

$$H_{ij}(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k} \cdot \mathbf{R}} \langle w_{i0} | \hat{H} | w_{j\mathbf{R}} \rangle$$

$w_{j\mathbf{R}}$: Wannier function j in the cell at $\mathbf{R}$; a few hundred $\mathbf{R}$ are enough.

7 → 1.44 M

graphene: DFT k-points → spin
Hall grid

24 → 0.59 M

MnTe: DFT k-points →
anomalous Hall grid

30 s, 1 min

the two integrals on an 8-core
laptop

Symmetry-adapted Wannier functions: DFT on irreducible k only

Each Wannier function should transform like its orbital:

$$\hat{g} w_{i\mathbf{R}}(\mathbf{r}) = \sum_j D_{ji}(g) w_{j,g\mathbf{R}}(\mathbf{r})$$

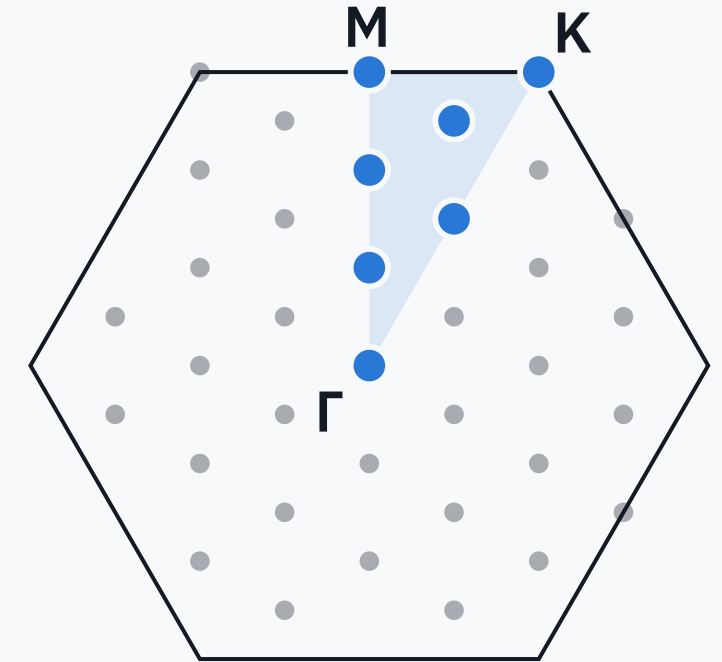
$\hat{g}$: symmetry operation, $D(g)$: its matrix on the orbitals of one site

Then the gauge at $g\mathbf{k}$ follows from $\mathbf{k}$ (Sakuma 2013):

$$U(g\mathbf{k}) = \tilde{d}(g, \mathbf{k}) U(\mathbf{k}) D(g, \mathbf{k})^\dagger$$

U : Bloch $\rightarrow$ Wannier gauge, $\tilde{d}$: g acting on the Bloch states (from irrep)

- DFT on 7 of 36 k-points for graphene, 24 of 144 for MnTe
- Degeneracies come out exact



6 × 6 grid: 36 points, 7 irreducible (blue)

Wannierisation: WannierBerri vs Wannier90

$$\Omega = \sum_n \left[\langle w_n | r^2 | w_n \rangle - \left| \langle w_n | \mathbf{r} | w_n \rangle \right|^2 \right] = \Omega_I + \tilde{\Omega}$$

Ω : spread of the Wannier functions · Ω_I : fixed by the chosen subspace (disentanglement) · $\tilde{\Omega}$: depends on the gauge $U(\mathbf{k})$ (localisation)

	Wannier90	WannierBerri, this demo
Input	.amn, .mmn, .eig files from a DFT interface	computed in Python from the GPAW wave functions
k-points	full grid; irreducible ones with site_symmetry	irreducible ones
Algorithm	disentangle first, then minimise $\tilde{\Omega}$ by steepest descent or conjugate gradients	one loop: a disentanglement step and a local gauge update per iteration, no gradient descent
Symmetry	site_symmetry with a .dmn file from pw2wannier90, not with a frozen window	from irrep, for any code it reads; frozen window allowed
SOC	spinor DFT run with SOC	collinear run, SOC added afterwards; spin ↓ from symmetry in altermagnets

Post-processing: WannierBerri vs postw90

$$H(\mathbf{K} + \boldsymbol{\kappa}) = \sum_{\mathbf{R}} e^{i\boldsymbol{\kappa} \cdot \mathbf{R}} \left[e^{i\mathbf{K} \cdot \mathbf{R}} H(\mathbf{R}) \right]$$

$\boldsymbol{\kappa}$: points of a coarse FFT grid · $\mathbf{K}$: shifts of that grid; one FFT per shift, the shifts together give the dense grid

	postw90	WannierBerri
$H(\mathbf{R}) \rightarrow H(\mathbf{k})$	sum over $\mathbf{R}$ at every $\mathbf{k}$, cost $\propto N_{\mathbf{R}}$	FFT over sub-grids, cost $\propto \log N_{\text{FFT}}$ per $\mathbf{k}$
Symmetry	full grid	irreducible points of the magnetic point group, result symmetrised
Hot spots	finer mesh where $ \Omega $ exceeds a threshold	recursive adaptive refinement
Fermi levels	cost grows with their number	~1000 at almost no extra cost
SOC axis	fixed by the DFT run	<code>set_soc_axis</code> : any axis and strength

Up to three orders of magnitude faster on dense grids. In this demo: 1401 Fermi levels in one run; for $\hat{\mathbf{n}} \parallel \mathbf{y}$ only the $\mathbf{k}$ -points irreducible under $m'm'm$ (8 operations) are computed.

Does graphene have a gap?

Step	Run	Checkpoint
0 · DFT	<code>python 1_dft.py</code>	10 s, <code>gpaw/nscf.gpw</code>
1 · Symmetry	irrep	48 operations, 7 of 36 k-points
2 · Wannier functions	<code>wannierise</code>	spread 0.91 Å ² each
3 · Bands	interpolation	blue on grey in the window
4 · SOC on demand	<code>set_soc_axis</code>	gap 35.8 μeV (GPAW 38.7)
5 · Spin Hall	<code>wb.run</code> , 1200 ² k	1.0003 e/2π, 30 s

Graphene: DFT in 10 s, two p_z Wannier functions

- PBE, 400 eV, $6 \times 6 \times 1$ grid, 15 Å vacuum, no SOC
- NSCF only on the 7 irreducible k-points
- Frozen window $E_F - 2.9 \dots E_F + 2.8$ eV, plus the π state at Γ

0.91 Å²

spread per function

48

symmetry operations

3 s + 2 s

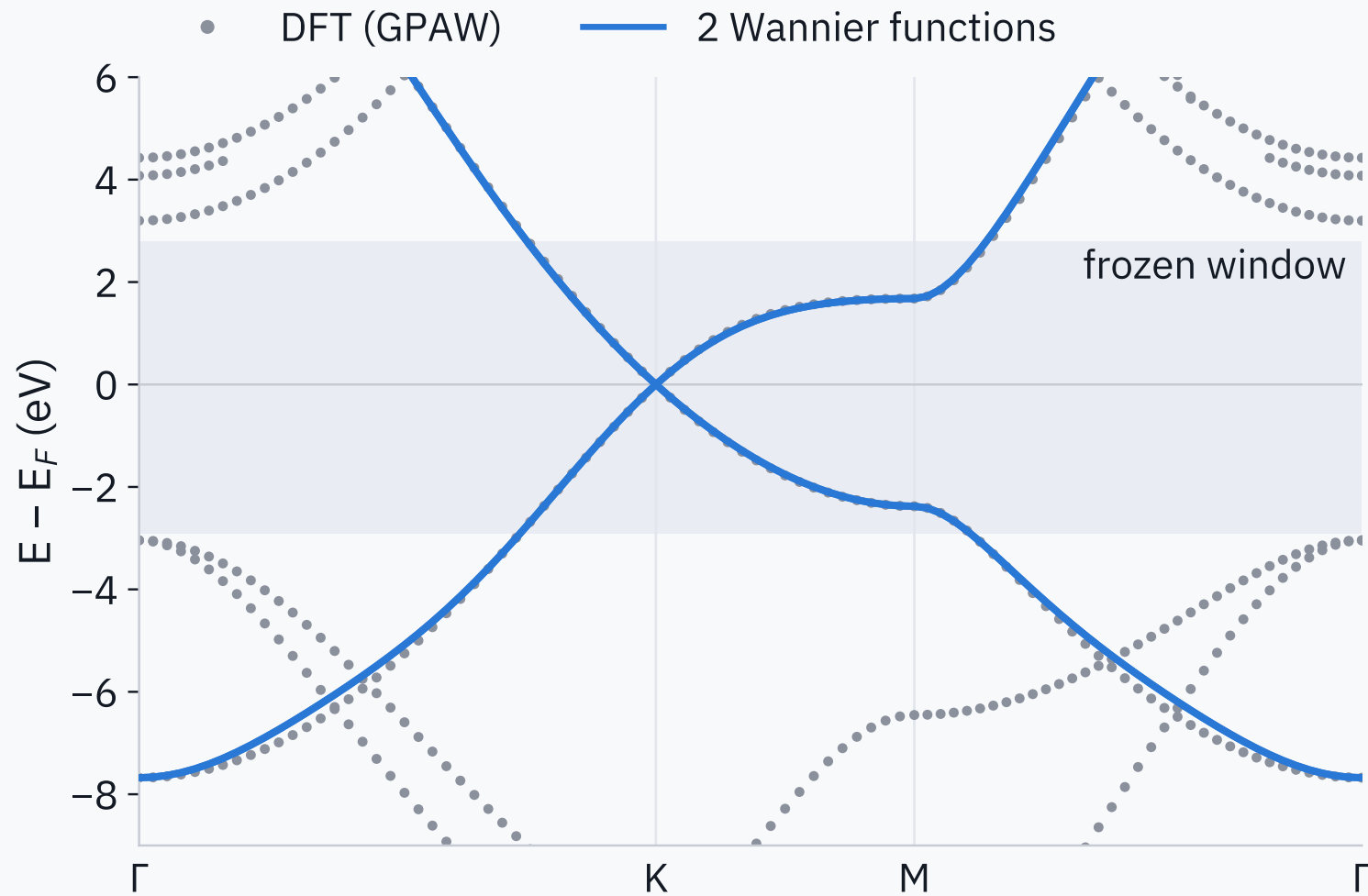
from_gpaw + wannierise

```
calc = GPAW(mode=PW(400), xc="PBE",
             kpts={"size": (6, 6, 1), "gamma": True})
atoms.calc = calc
atoms.get_potential_energy()
sg = SpaceGroup.from_gpaw(calc) # irrep
kpts = sg.get_irreducible_kpoints_grid((6, 6, 1)) # 7 of 36
calc.fixed_density(kpts=kpts, nbands=20).write(
    "gpaw/nscf.gpw", mode="all")

# graphene.ipynb
pz = Projection(position_num=[[1/3, 2/3, 0], [2/3, 1/3, 0]],
                orbital="pz", spacegroup=sg)
wandata = wb.WannierDataSOC.from_gpaw(
    calc, projections=ProjectionsSet([pz]))
wandata.wannierise(froz_min=EF - 2.9, froz_max=EF + 2.8,
                  frozen_states={0: [1]}, # pi at Gamma
                  num_iter=100, sitesym=True)
system = wb.SystemSOC.from_wannierdata(wandata, berry=True)
```

Two Wannier bands follow DFT through the frozen window

- Grey: GPAW along Γ –K–M– Γ , not used in the fit
- Blue: the two-band Wannier model
- Shaded: frozen window



SOC as a matrix you can rotate and scale

$$H(\mathbf{k}) = H_0(\mathbf{k}) \otimes \sigma_0 + \alpha \sum_c \Delta V^c(\mathbf{k}) \otimes \sigma_c(\hat{\mathbf{n}})$$

$$\Delta V_{mn}^c(\mathbf{k}) = \sum_{a,ij} \langle \tilde{\psi}_{m\mathbf{k}} | \tilde{p}_i^a \rangle \langle \phi_i^a | \xi_a(r) L_c | \phi_j^a \rangle \langle \tilde{p}_j^a | \tilde{\psi}_{n\mathbf{k}} \rangle$$

$$\xi_a(r) = \frac{1}{4m^2c^2} \frac{1}{r} \frac{dV_a}{dr}$$

H_0 : Wannier Hamiltonian without SOC · $\sigma_c(\hat{\mathbf{n}})$: Pauli matrices, z along the spin axis $\hat{\mathbf{n}}$ · α : SOC scale, 1 = real · $\tilde{\psi}, \tilde{p}_i^a, \phi_i^a$: PAW pseudo wave functions, projectors, partial waves of atom a · L_c : orbital angular momentum · V_a : potential in sphere a

- Same PAW matrix elements as GPAW's own SOC
- New $\hat{\mathbf{n}}$ or α : no new DFT

```
K = [1/3, 1/3, 0]
energy = wb.calculators.tabulate.Energy(
    degen_thresh=1e-8)
for alpha in [0, 1]:
    system.set_soc_axis(alpha_soc=alpha)
    E = wb.evaluate_k(system, k=K,
        calculators={"E": energy})
```

	E(K) - E_F (ueV)				gap
alpha = 0	28.6	28.6	28.6	28.6	0.0
alpha = 1	10.7	10.7	46.5	46.5	35.8
GPAW SOC	9.0	9.0	47.7	47.7	38.7

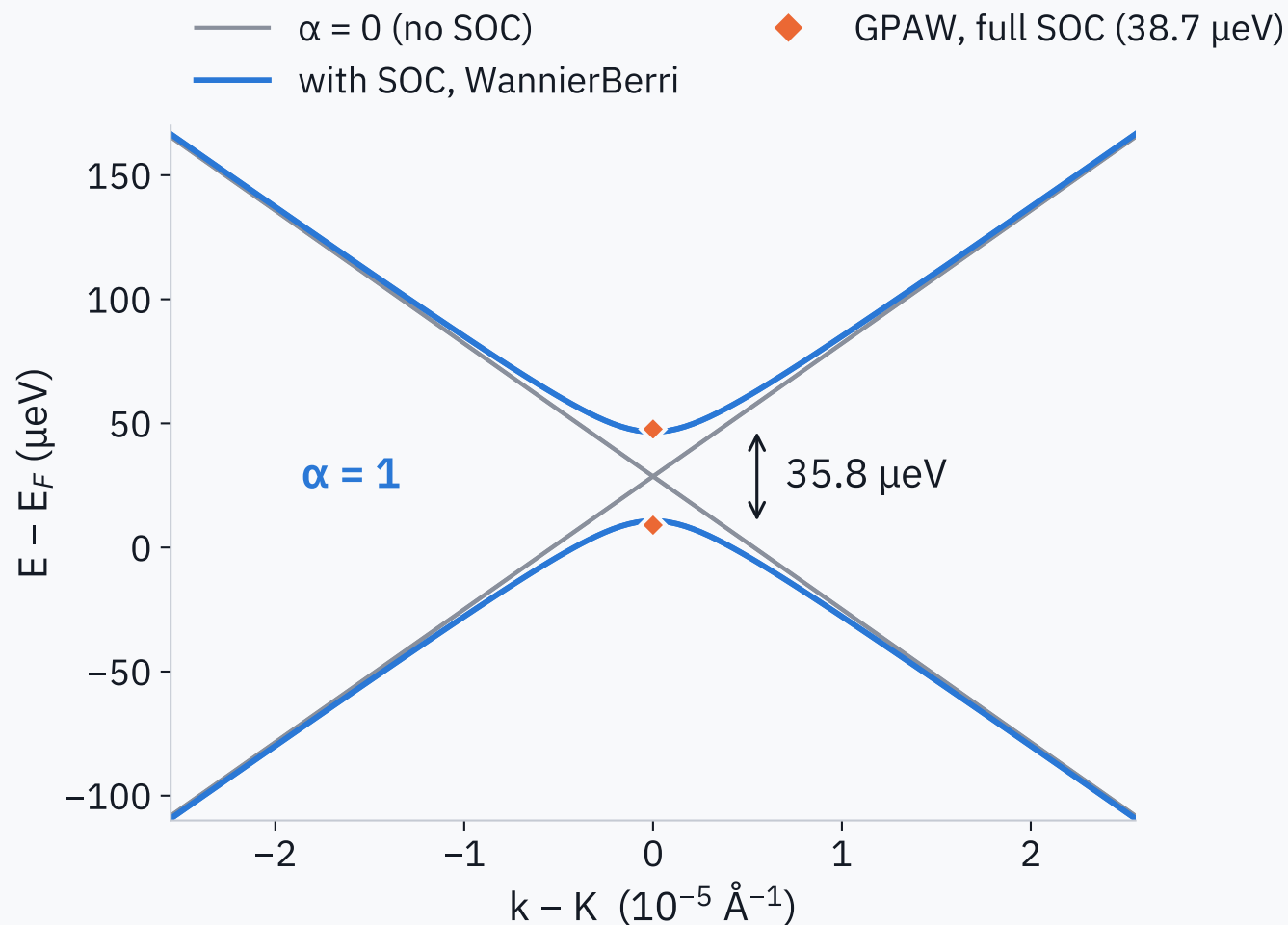
`degens_thresh=1e-8`: the default 10^{-4} eV would merge the split pair

Kane–Mele gap: 36 μeV from two Wannier functions

Gap at K: **35.8 μeV**

GPAW, full SOC, 20 bands: **38.7 μeV**

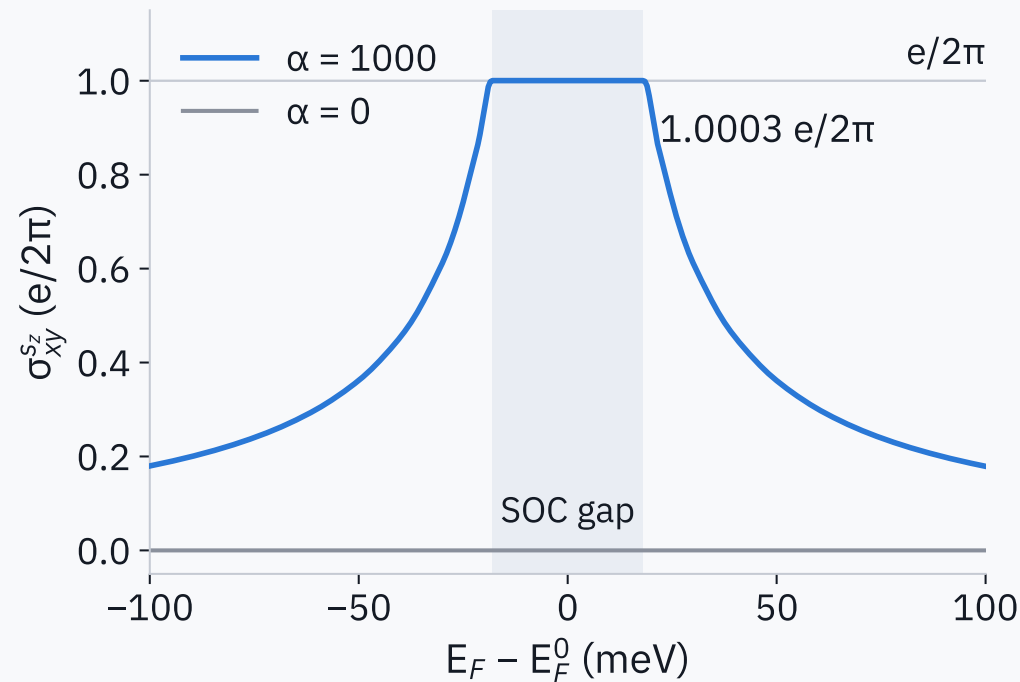
- p_z has $m = 0$, so the gap comes from the small d part of p_z in the PAW spheres (Gmitra 2009)
- Missing 3 μeV : second-order SOC via the σ bands
- Depends on cutoff and PAW dataset; experiment 42.2 μeV (Sichau 2019)



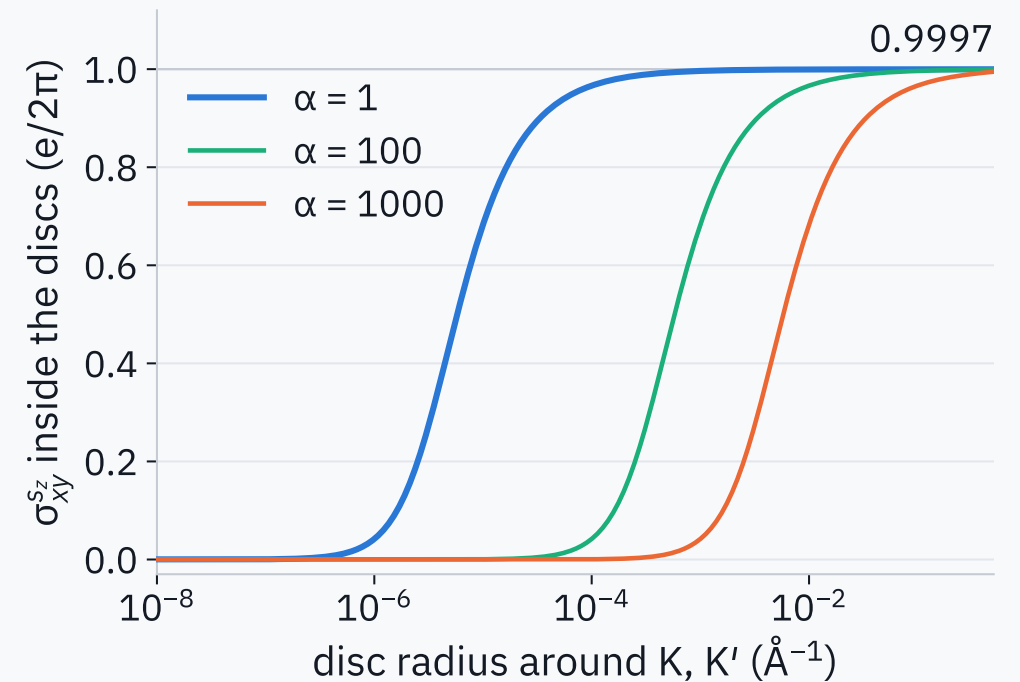
Spin Hall conductivity: $e/2\pi$, also at the real gap

$$\sigma_{xy}^{sz} = \frac{e}{\hbar} \int \frac{d^2k}{(2\pi)^2} \sum_n f_{n\mathbf{k}} \Omega_n^{sz}, \quad \Omega_n^{sz} = -2\hbar^2 \operatorname{Im} \sum_{m \neq n} \frac{(j_x^{sz})_{nm} (v_y)_{mn}}{(\varepsilon_n - \varepsilon_m)^2}$$

$j_x^{sz} = \frac{1}{2} \{s_z, v_x\}$: spin current, v : velocity. With s_z conserved, $\sigma_{xy}^{sz} = (e/2\pi)(C_\uparrow - C_\downarrow)/2 = e/2\pi$



$\alpha = 1000$, 1200^2 grid with tetrahedra: $1.0003 e/2\pi$



$\alpha = 1$ (36 μeV), log-polar grid around K and K': $0.9997 e/2\pi$

A Hall effect without magnetisation

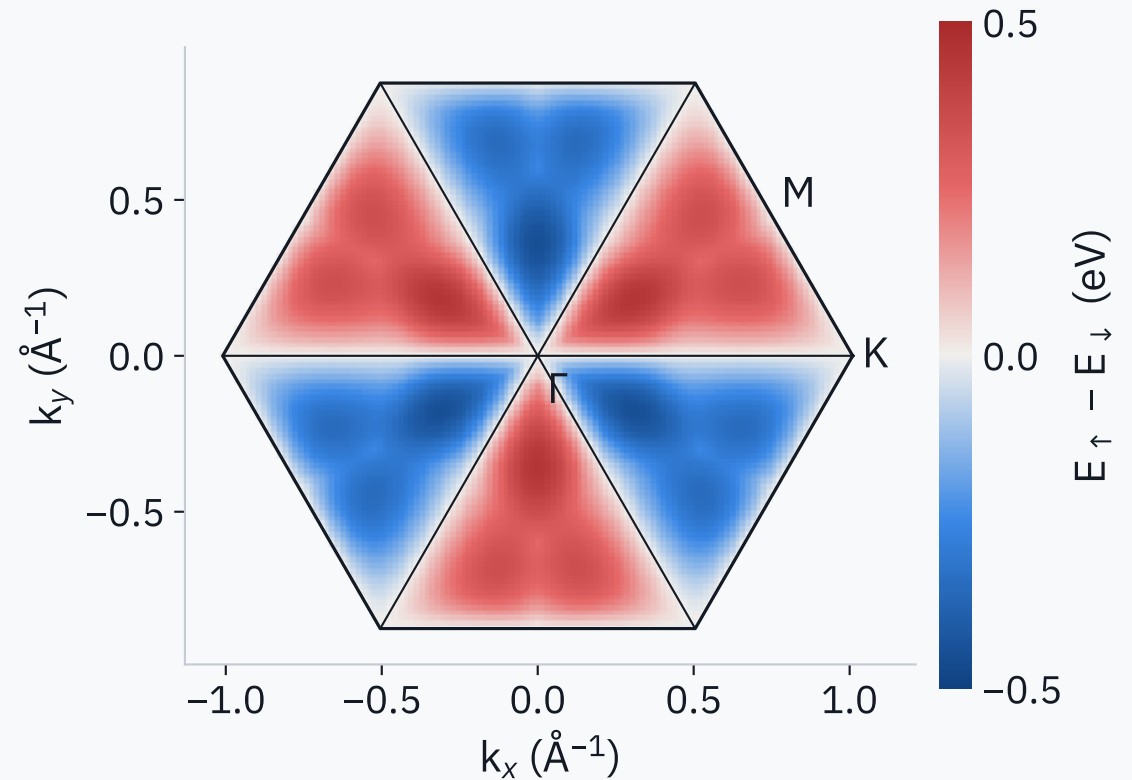
Step	Run	Checkpoint
0 · DFT	<code>python 1_dft.py</code>	45 s, moments $\pm 4.41 \mu_B$
1–2 · Spin $\uparrow$ Wannier	<code>altermagnetic=True</code>	24 of 144 k-points, 25 s
3 · Splitting	no SOC	up to 0.97 eV at $k_z \neq 0$
4 · Turn $\hat{n}$	<code>set_soc_axis</code>	shift at A: 322 $\rightarrow$ 75 meV
5 · Anomalous Hall	three $\hat{n}$	0 for x and c, up to 194 S/cm for y
6 · Rotate in plane	0–30°	$\sigma(5^\circ)/\sigma(30^\circ) = 0.56$

Altermagnets: zero net moment, spin-split bands without SOC

- Two sublattices related by a rotation or mirror, not by a translation or inversion
- Bands spin split without SOC; the sign alternates in $\mathbf{k}$ (d-, g-, i-wave)
- No PT or tT symmetry, so with SOC an anomalous Hall effect is possible, depending on the Néel vector
- **MnTe**: NiAs type, $T_N \approx 310$ K, gap 1.27 eV

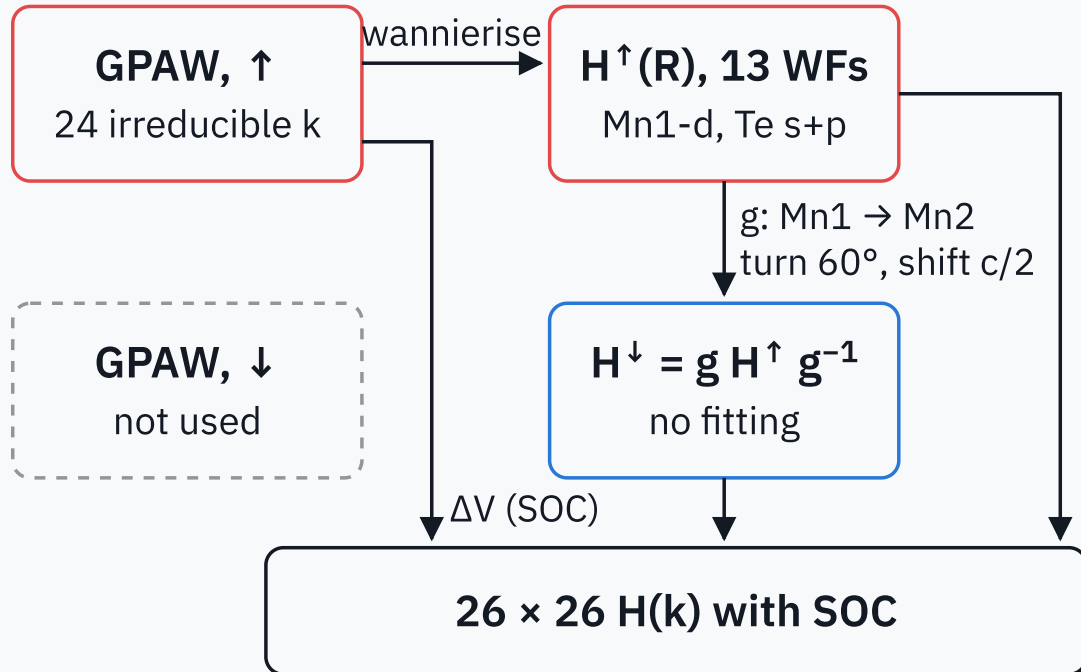
$$[C_2 \parallel g] : E_{\uparrow}(\mathbf{k}) = E_{\downarrow}(R_g \mathbf{k})$$

C_2 : flips all spins, g : swaps the Mn sublattices, R_g : its rotation part



MnTe, no SOC: $E_{\uparrow} - E_{\downarrow}$ at $k_z = 0.35 \text{ \AA}^{-1}$

Wannierise spin $\uparrow$ only, spin $\downarrow$ follows by symmetry



PBE+U ($U = 4$ eV), $6 \times 6 \times 4$, 45 s: moments $\pm 4.41 \mu_B$, gap 0.77 eV (exp. 1.27–1.41 eV; a larger U does not fix it)

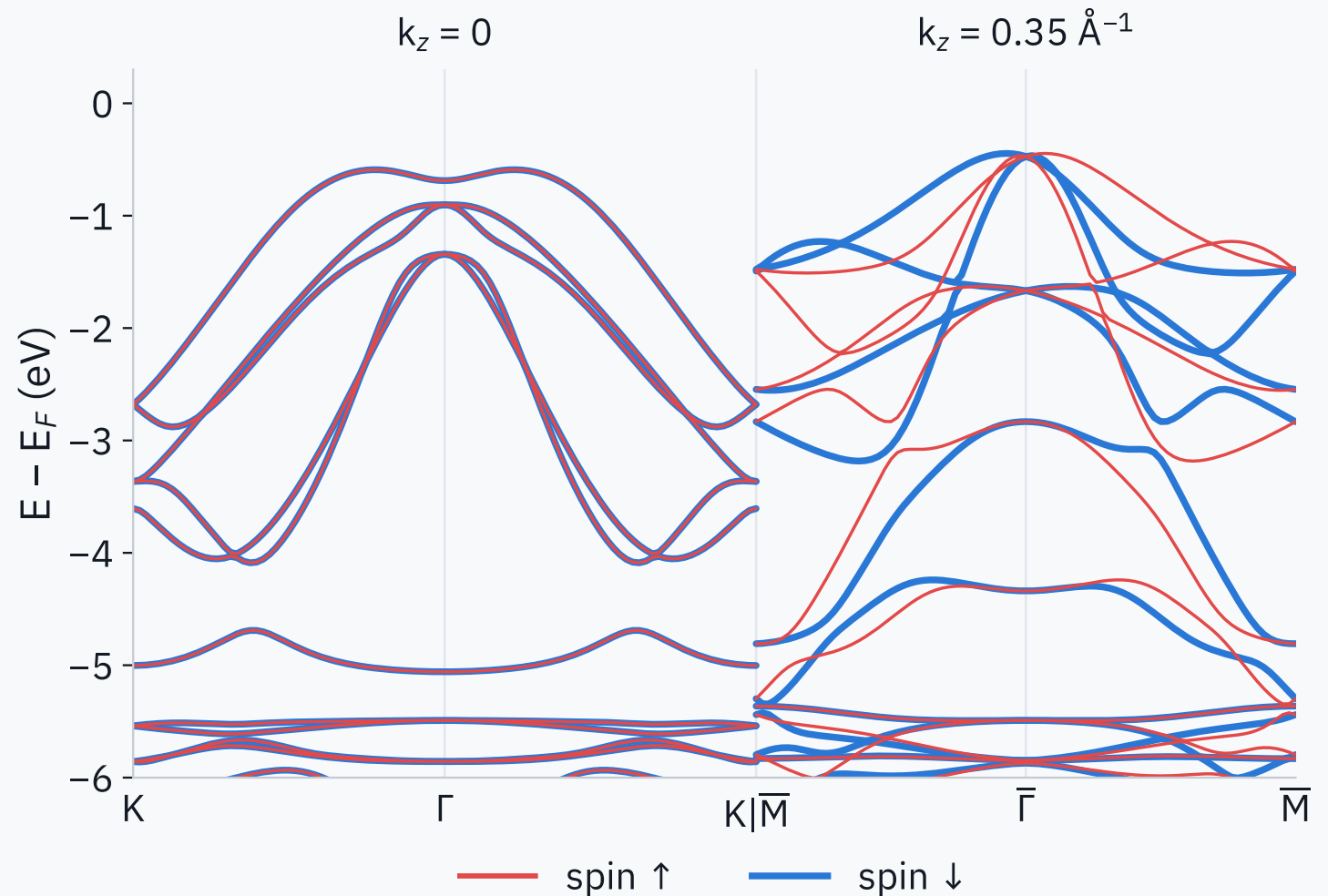
```

calc = GPAW("gpaw/nscf.gpw")    # PBE+U, 24 of 144 k
sg = SpaceGroup.from_gpaw(calc)
Te = [[1/3, 2/3, 1/4], [2/3, 1/3, 3/4]]
projections = ProjectionsSet([
    Projection(position_num=[0, 0, 0], orbital="d",
               spacegroup=sg),          # Mn1 only
    Projection(position_num=Te, orbital="sp2",
               spacegroup=sg, xaxis=[0, -1, 0],
               rotate_basis=True),
    Projection(position_num=Te, orbital="pz",
               spacegroup=sg)])

wandata = wb.WannierDataSOC.from_gpaw(
    calc, projections=projections,
    altermagnetic=True,    # spin down from symmetry
    IBstart=8)             # skip Mn 3s, 3p
wandata.wannierise(froz_min=-np.inf,
                  froz_max=EF + 0.1,
                  num_iter=200, sitesym=True)
system = wb.SystemSOC.from_wannierdata(wandata,
                                       berry=True)
    
```

No SOC: degenerate at $k_z = 0$, split by up to 1 eV at $k_z \neq 0$

- Red: spin $\uparrow$, Wannierised
- Blue: spin $\downarrow$, from symmetry
- At $k_z = 0.35 \text{ \AA}^{-1}$ up to 0.97 eV apart, with opposite order on the two sides of Γ : g-wave



With SOC the Néel vector picks the magnetic point group

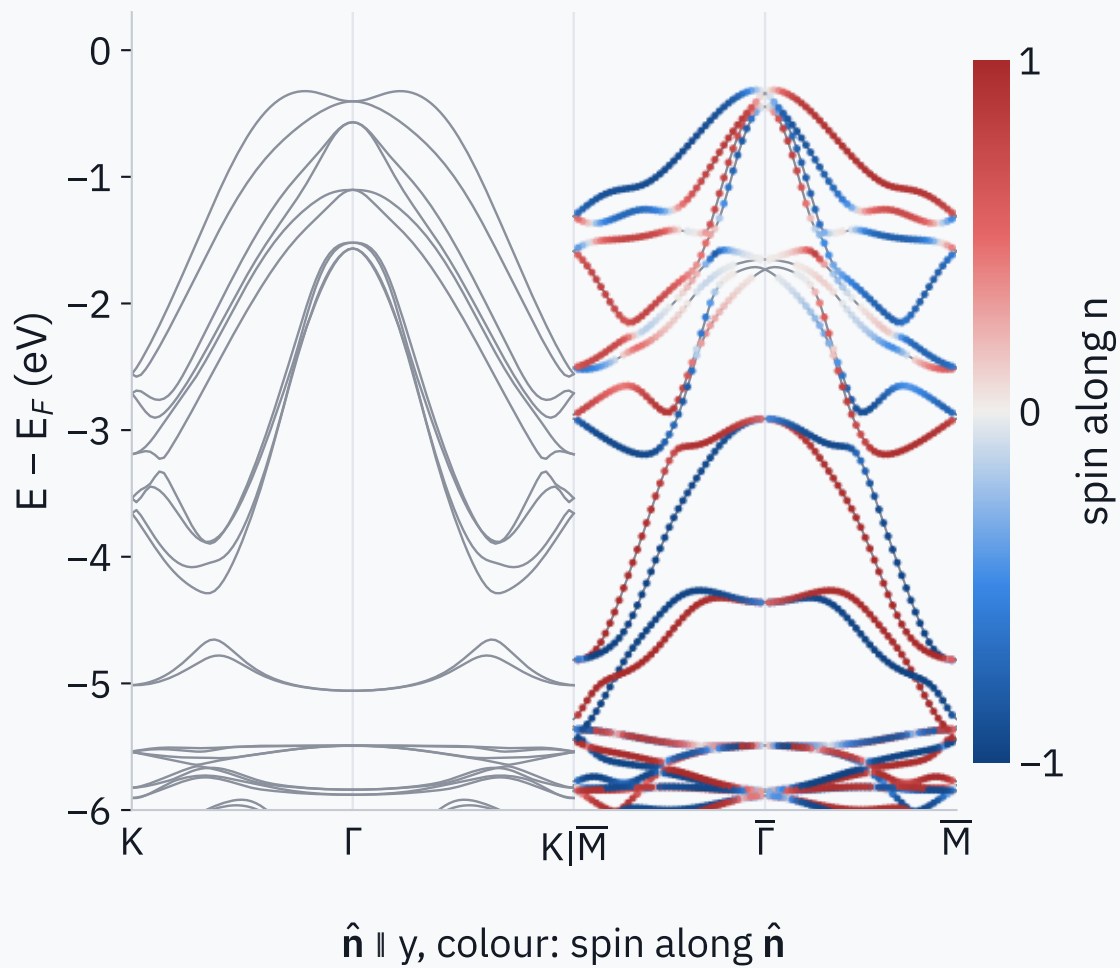
Néel vector $\hat{n}$	Group	Hall vector
$\parallel c$	$6'/m'm'm$	none
$\parallel a = x$	mmm	none
$\perp a = y$	$m'm'm$	$\sigma_{xy} \neq 0$

$$g : \sigma_H \rightarrow \det(R_g) R_g \sigma_H$$

$$\mathcal{T} : \sigma_H \rightarrow -\sigma_H$$

$\sigma_H = (\sigma_{yz}, \sigma_{zx}, \sigma_{xy})$: Hall vector · R_g : rotation part of g · $\mathcal{T}$: time reversal · primed: combined with $\mathcal{T}$

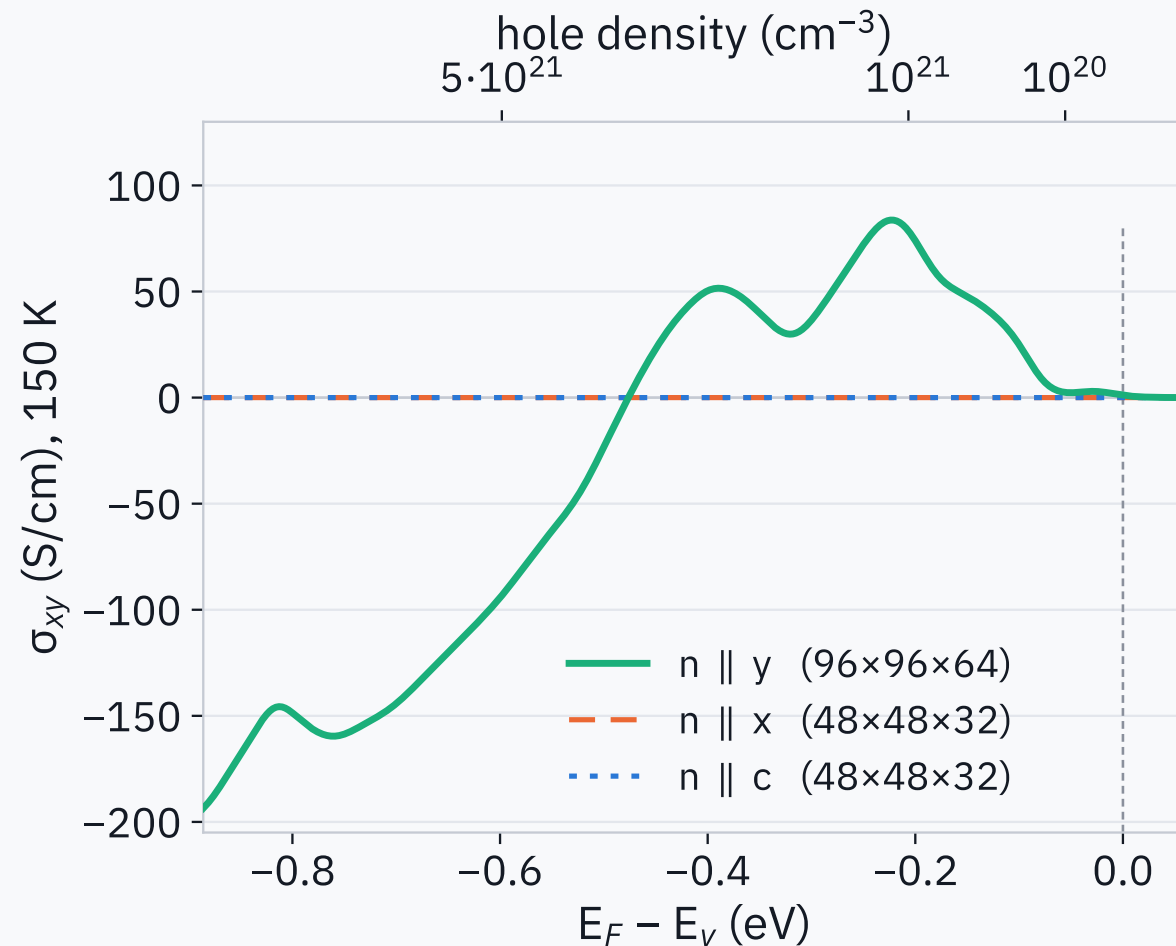
SOC shift of the top level at A, model vs GPAW's full SOC: 322 vs 321 meV ($\hat{n} \parallel c$), 75 vs 60 meV ($\hat{n}$ in plane)



Anomalous Hall conductivity for three Néel vectors

```
ahc = wb.calculators.static.AHC(Efermi=Efermi,
    degen_thresh=1e-4, smoother=smoother)
directions = {"x": (90, 0), "y": (90, 90),
              "z": (0, 0)} # theta, phi
grids = {"x": (48, 48, 32), "y": (96, 96, 64),
         "z": (48, 48, 32)}
for name, (theta, phi) in directions.items():
    system.set_soc_axis(theta=theta, phi=phi,
                        units="degrees")
    result = wb.run(system,
                    grid=wb.Grid(system, NK=grids[name]),
                    calculators={"ahc": ahc})
```

- $\hat{n} \parallel x$ and $\hat{n} \parallel c$: zero, as symmetry says
- $\hat{n} \parallel y$: up to 84 S/cm, sign change 0.48 eV below the valence band top E_v



Symmetry gives $\sin 3\varphi$

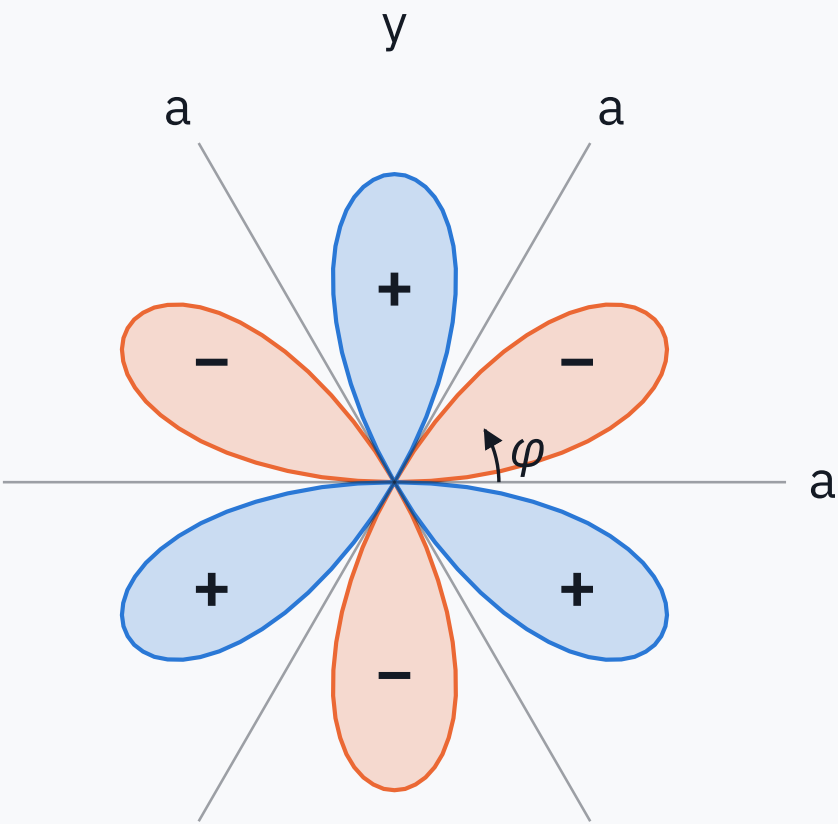
In-plane Néel vector at angle φ from a:

Operation	Swaps Mn	$\varphi \rightarrow$	So
turn 180° about a	no	$-\varphi$	$\sigma(-\varphi) = -\sigma(\varphi)$
turn 60° about c, shift $c/2$, then $\mathcal{T}$	yes	$\varphi + 60^\circ$	$\sigma(\varphi + 60^\circ) = -\sigma(\varphi)$

$$\sigma_{xy}(\varphi) = A_3 \sin 3\varphi + A_9 \sin 9\varphi + A_{15} \sin 15\varphi + \dots$$

A_m : amplitudes in S/cm

- Fixed by symmetry: zeros along every a axis, sign change every 60°
- Not fixed: the A_m ; $\sin 3\varphi$ need not dominate



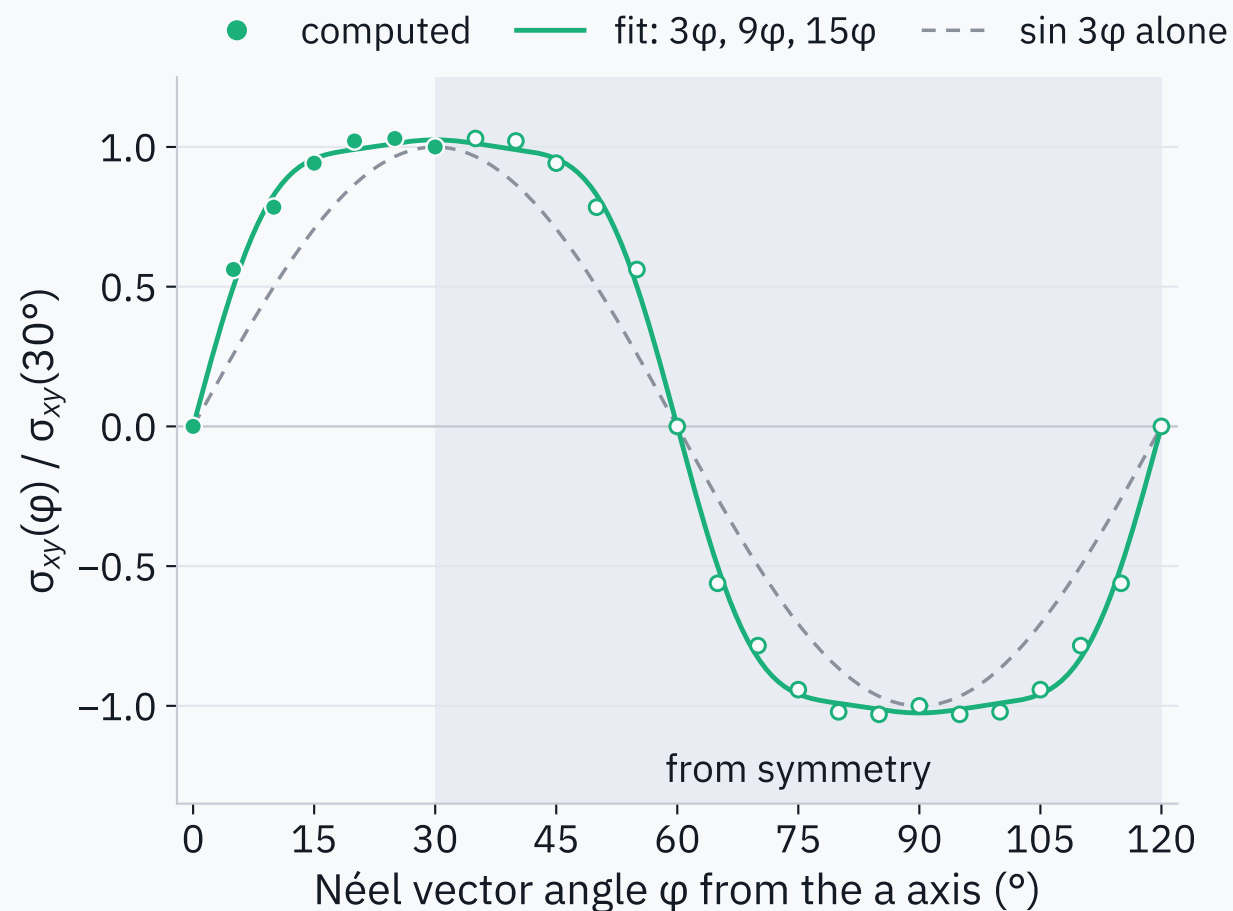
Sign of σ_{xy} against the direction of $\hat{n}$, $\sin 3\varphi$ only (sketch)

Rotating the Néel vector: exact zeros, but not a $\sin 3\varphi$ curve

One `set_soc_axis` per angle, no new DFT: 0–30° in 5° steps, 1 min

- $\sigma(-\varphi) = -\sigma(\varphi)$ and $\sigma(\varphi + 60^\circ) = -\sigma(\varphi)$ hold to $3 \cdot 10^{-7}$ S/cm
- $\sigma(5^\circ)/\sigma(30^\circ) = 0.56$, $\sin 3\varphi$ alone gives 0.26
- Fit: $A_3 = 172$, $A_9 = 30$, $A_{15} = 7$ S/cm

$E_F = E_V - 0.7$ eV ($6.6 \cdot 10^{21}$ holes/cm³), 48×48×32 grid



Filled: computed. Open: from the two symmetry relations

Your material next

```
calc = GPAW(...) # SCF
kpts = SpaceGroup.from_gpaw(calc)\
    .get_irreducible_kpoints_grid(grid)
calc.fixed_density(kpts=kpts, nbands=...) # NSCF
wandata = wb.WannierDataSOC.from_gpaw(
    calc, projections=...)
wandata.wannierise(froz_min=..., froz_max=...,
    sitesym=True)
system = wb.SystemSOC.from_wannierdata(
    wandata, berry=True)
system.set_soc_axis(theta, phi)
wb.run(system, grid=..., calculators=...)
```

- Projections: the orbitals that make the bands you care about
- Frozen window: every state you need kept exactly
- k-grids: multiples of the DFT grid
- `degen_thresh` below the smallest splitting
- Check a few levels against GPAW first

Take-aways, and how to run it

- Symmetry: DFT on 7 of 36 and 24 of 144 k-points
- SOC on demand: any spin axis and strength, no new DFT
- Spin Hall plateau $e/2\pi$ at the real gap
- Altermagnets: Wannierise one spin channel, get both
- The Néel vector is just a parameter: symmetry gives the zeros, the bands give the size

RUN IT

github.com/yaroslav-zhumagulov/wannier-berri-demo

```
bash setup/setup.sh      # once, 15 min
conda activate wbdemo
bash run_all.sh           # 6 min, or the notebooks
```

Linux, macOS or WSL2, with GPAW 26.7 from conda-forge

BACKUP

Details

Kane–Mele form · nodal planes of MnTe · uniform grids · timings ·
references

The Wannier SOC matrix has the Kane–Mele form

$$H_{\text{SO}} = i\lambda \sum_{\langle\langle ij \rangle\rangle} \nu_{ij} c_i^\dagger s_z c_j, \quad \Delta_{\text{SO}} = 6\sqrt{3} \lambda$$

- Second neighbours: $\lambda = 3.43 \text{ } \mu\text{eV}$, purely imaginary, sign ± 1 as in Kane–Mele
- $6\sqrt{3} \lambda = 35.6 \text{ } \mu\text{eV}$, full model $35.8 \text{ } \mu\text{eV}$
- $\Delta V^{x,y} = 0$: the mirror plane forbids Rashba terms, s_z is conserved
- No SOC between first neighbours: each bond lies in a vertical mirror

Shell	$ d \text{ (}\text{\AA}\text{)}$	$ t \text{ (eV)}$	$ \Delta V^{x,y} \text{ (eV)}$	$ \Delta V^z \text{ (}\mu\text{eV}\text{)}$
1st, A–B	1.42	2.90	0	0
2nd, A–A	2.46	0.25	10^{-25}	3.43
3rd, A–B	2.84	0.26	10^{-26}	0
4th, A–B	3.76	0.02	10^{-26}	0
5th, A–A	4.26	0.04	10^{-25}	0
6th, A–A	4.92	0.03	10^{-25}	0.26

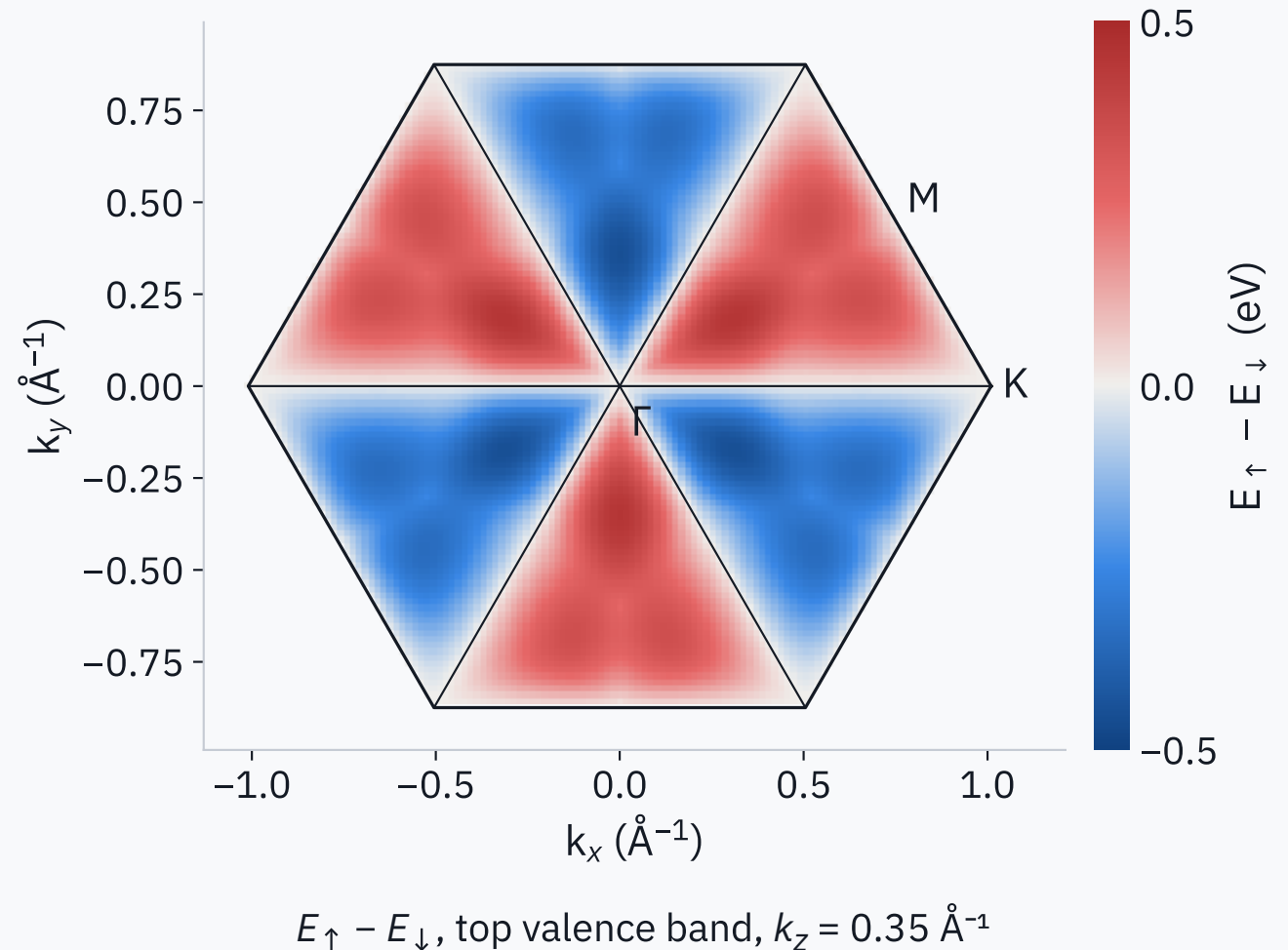
Largest element per shell of $H(\mathbf{R})$ and $\Delta V(\mathbf{R})$; nothing fitted

Four nodal planes make MnTe g-wave

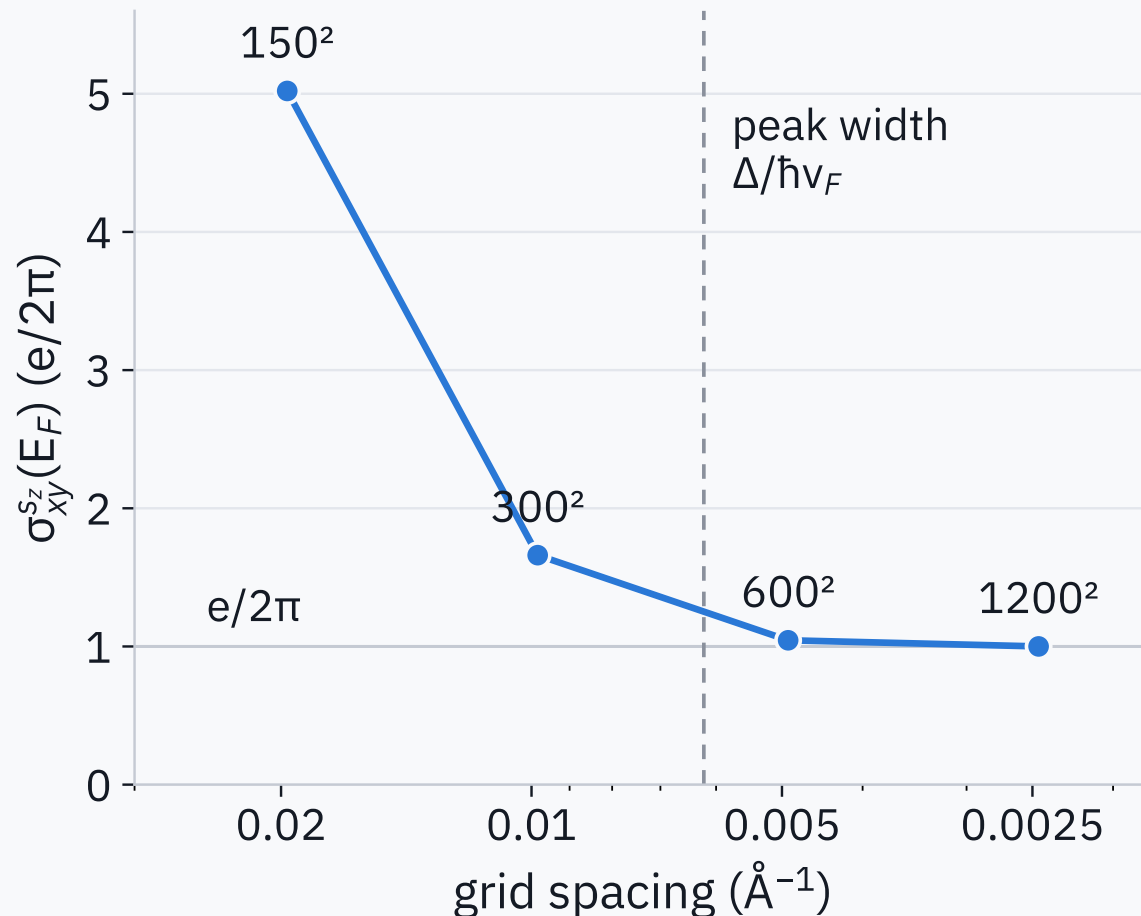
$E_{\uparrow}(\mathbf{k}) = E_{\downarrow}(R_g \mathbf{k})$ for a spin-flipping g , so the splitting vanishes on planes that R_g leaves in place:

- m_z : $k_z = 0$ and $k_z = \pi/c$
- three c -glides: three vertical planes through Γ –K

In the model: $|E_{\uparrow} - E_{\downarrow}| \leq 1.5 \cdot 10^{-14}$ eV on these planes, up to 0.95 eV along Γ –M at $k_z = 0.35 \text{ \AA}^{-1}$



Uniform grids overshoot until they resolve the peak at K



$\alpha = 1000$ (gap 36 meV), plain k-sum at E_F , 4 s for all four grids

- Spin Berry curvature: a peak at K, width $\Delta/\hbar v_F \approx 0.006 \text{ \AA}^{-1}$
- K is on every grid divisible by 6, so coarse grids give the peak too much weight
- 1200^2 : 1.0003 $e/2\pi$
- Outside the gap, tetrahedra remove the steps of a plain sum

Where the time goes

Step	Graphene	MnTe
GPAW SCF + NSCF, one core	10 s	45 s
<code>WannierDataSOC.from_gpaw</code>	3.3 s	9.7 s
<code>wannierise</code>	2.1 s	9.7 s
<code>SystemSOC.from_wannierdata</code>	1.0 s	6.6 s
Conductivities (step 5)	28 s	63 s
Néel scan (step 6)	–	66 s
Animation frames	1 s	22 s
Whole notebook	1.1 min	4.2 min

Apple M2, 8 cores, 8 Ray workers:
6 minutes for everything, try-its
and bonus steps included

With the Wannier model in hand, a
new spin axis, SOC strength or
Fermi level costs seconds

References

CODES

- WannierBerri: S. S. Tsirkin, npj Comput. Mater. **7**, 33 (2021)
- GPAW: J. J. Mortensen et al., J. Chem. Phys. **160**, 092503 (2024)
- IrRep: M. Iraola et al., Comput. Phys. Commun. **272**, 108226 (2022)
- Wannier90: G. Pizzi et al., J. Phys.: Condens. Matter **32**, 165902 (2020)
- Wannier ecosystem: A. Marrazzo et al., Rev. Mod. Phys. **96**, 045008 (2024)

METHODS

- N. Marzari, D. Vanderbilt, PRB **56**, 12847 (1997); I. Souza et al., PRB **65**, 035109 (2001); N. Marzari et al., Rev. Mod. Phys. **84**, 1419 (2012)
- R. Sakuma, PRB **87**, 235109 (2013)
- D. Xiao, M.-C. Chang, Q. Niu, Rev. Mod. Phys. **82**, 1959 (2010)
- X. Wang et al., PRB **74**, 195118 (2006); J. Qiao et al., PRB **98**, 214402 (2018); J. H. Ryoo et al., PRB **99**, 235113 (2019)

PAW, SOC AND DFT+U

- P. E. Blöchl, PRB **50**, 17953 (1994); T. Olsen, PRB **94**, 235106 (2016); S. L. Dudarev et al., PRB **57**, 1505 (1998)

GRAPHENE

- C. L. Kane, E. J. Mele, PRL **95**, 226801 and 146802 (2005)
- M. Gmitra et al., PRB **80**, 235431 (2009); J. C. Boettger, S. B. Trickey, PRB **75**, 121402 (2007)
- J. Sichau et al., PRL **122**, 046403 (2019)

ALTERMAGNETS AND MNTE

- L. Šmejkal et al., Sci. Adv. **6**, eaaz8809 (2020); PRX **12**, 031042 and 040501 (2022)
- D. Kriegner et al., PRB **96**, 214418 (2017); D. Bossini et al., New J. Phys. **22**, 083029 (2020)
- R. D. Gonzalez Betancourt et al., PRL **130**, 036702 (2023)
- J. Krempaský et al., Nature **626**, 517 (2024)